

BEYOND SHAPEFILE: A TAXONOMY OF MODERN GEOSPATIAL VECTOR DATA FORMATS

Markus Tremmel
FOSS4G Hiroshima 2026

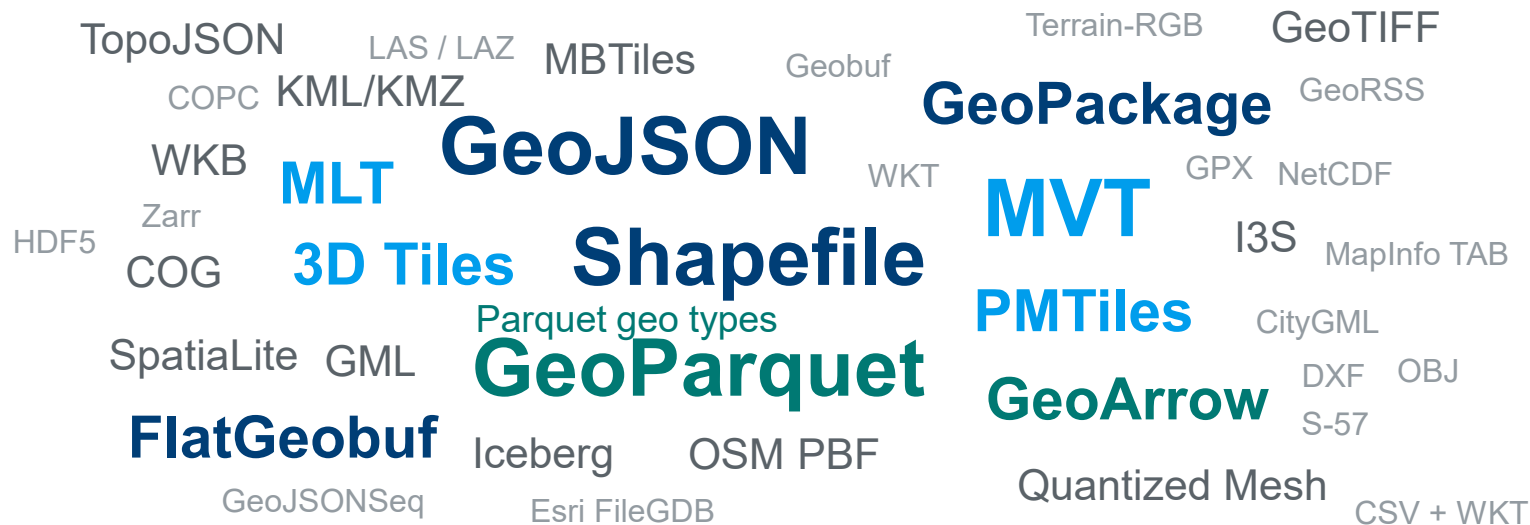
ROHDE & SCHWARZ

Make ideas real



FORMAT LANDSCAPE

GDAL alone reads **80+ vector** and **100+ raster** formats



Which format for which job?

High-performance maps.



Industry

Staff Software Engineer & Architect at Rohde & Schwarz, focused on high-performance map & geospatial applications



Open Source

Creator of the MapLibre Tile (MLT) and COMTiles formats



Academia

PhD research on the next-gen (cloud- & GPU-native) rendering stack · teaching vector data processing at Deggendorf Institute of Technology



AGENDA

1 · Data model

One logical model
behind (nearly)
every format

2 · Taxonomy

Three families,
six design axes

3 · Interchange Formats

Shapefile, GeoJSON,
GML, GeoPackage,
FlatGeobuf

4 · Visualization Formats

MVT, MLT,
PMTiles,
3D Tiles, I3S

5 · Analytics Formats

GeoPackage,
GeoParquet,
GeoArrow

6 · What's Next

Immersive 3D Map Views,
Gaussian Splatting,
Spatial Embeddings, ...

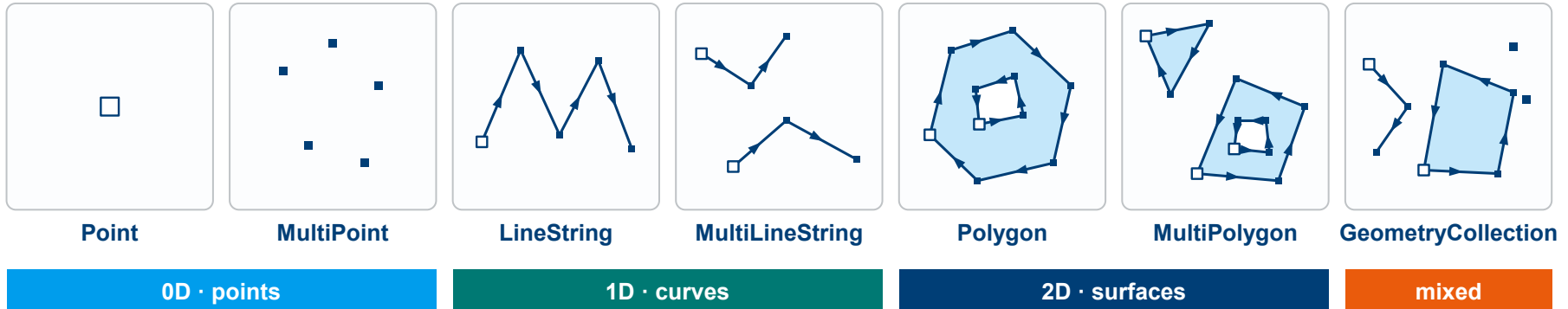
Scope: vector data. Raster gets a mention where it shares machinery (terrain).

ONE LOGICAL MODEL

- **Logical model: FeatureCollection** → **Feature** → **Geometry + Attributes**
- The geometry model used by nearly all vector formats follows the **OGC Simple Feature Access** standard: **Point**, **LineString**, **Polygon**, and their **Multi-variants**
- More complex geometry models, such as the **ISO Spatial Schema**, add support for non-linear interpolation between vertices, including **splines** and **Bézier curves**, but are rarely used in the GIS domain

```
{
  "type": "FeatureCollection",
  "features": [
    {
      "type": "Feature",
      "properties": {
        "name": "Marienplatz"
      },
      "geometry": {
        "type": "Point",
        "coordinates": [
          11.5758159,
          48.1371059
        ]
      }
    }
  ]
}
```

OGC Simple Feature Access Geometry Model



SAME LOGICAL MODEL → DIFFERENT PHYSICAL MODELS

1 · Storage layout

- row: feature after feature
- column: all values of one attribute
- hybrid (PAX): row groups/vectors of column chunks

2 · Structural encoding

how nesting and null values are encoded:

- offset/length + validity (Arrow, MLT)
- repetition / definition levels (Dremel → Parquet)

3 · Encodings & compression

- none, or general-purpose (Gzip, Zstd, ...)
- lightweight endings: quantisation, delta, varint, zigzag, rle

4 · Chunking

- one monolithic blob
- row groups, pages, tiles → streamable

5 · Indexing

- none → scan everything
- R-tree, KD-tree, Quadtree, Octree, zone maps, bloom filter → predicate and projection pushdown

6 · Serialization

text	parse every character
binary	parse fixed-width records
zero-copy	no deserialization at all

Every format is a particular set of choices along these six axes

A TAXONOMY OF GEOSPATIAL VECTOR FORMATS

1 · Interchange

Interoperability

Text

GeoJSON · GML · KML · TopoJSON

Binary

Shapefile · FlatGeobuf

Single-file container

GeoPackage

2 · Visualization

Render optimized → Tile-based

2D and 2.5D maps

MVT · MLT

Terrain

Quantized Mesh · Terrain-RGB

3D models

3D Tiles · I3S

Tile containers

MBTiles · PMTiles

3 · Analytics

Analytics Workflows

Transactional · OLTP

GeoPackage · SpatiaLite

Columnar on disk · OLAP

GeoParquet · Iceberg

Columnar in memory · OLAP

GeoArrow

Every format is listed under its **primary use case**. Raster-encoded terrain tiles are included because of their role in render-optimized terrain visualization.

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THE SHAPEFILE ERA AND ITS LIMITS

The Legacy: Interchange Formats

- ▶ **Shapefile, GML, GeoJSON**
- ▶ Built for broad interoperability and data exchange
- ▶ Deeply integrated into GIS desktop tools and workflows
- ▶ **But:** Text formats (GML, GeoJSON) have huge parsing overhead
- ▶ **But:** Binary formats (Shapefile) lack modern compression
- ▶ **But:** No internal chunking or HTTP range-read support (FlatGeobuf is the exception)

The Modern Demands

- ▶ Planet-scale basemaps at 60 FPS
- ▶ Analytics on massive datasets (TB/PB scale)
- ▶ Disaggregated cloud architecture (S3/R2)
- ▶ GPU-driven rendering

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One format can't do it all — the era of workload-specific formats

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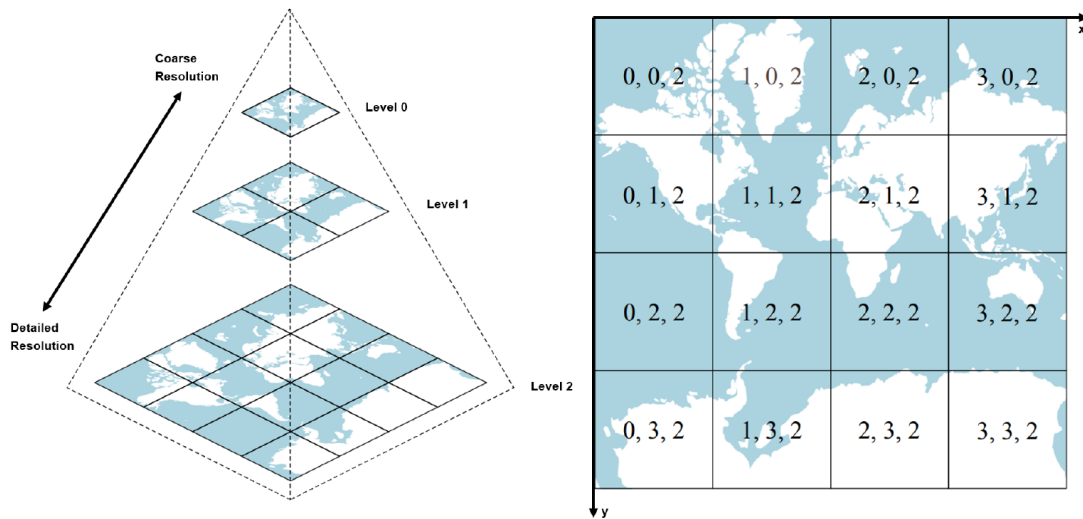
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VECTOR TILES

- ▶ Vector tiles are optimized for high-performance map rendering, particularly for large-scale basemap visualization based on the following concepts
 - tile-based architecture
 - multi-resolution overviews (tile pyramid) → cartographic generalization
 - coordinate quantization to integer grids (e.g., 4096×4096)
 - attribute pruning



Extended from Ammann et al. 2022

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2.5D rendering of LOD1 buildings using MapLibre GL JS



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- ▶ **Mapbox Vector Tile (MVT)**: The current industry standard for vector tile delivery, enabling 2.5D visualization
 - row-oriented storage layout
 - relies heavily on general-purpose compression (Gzip, Zstd, ...)
- ▶ **MapLibre Tile (MLT)**: The columnar successor designed for more efficient storage and decoding
 - column-oriented layout combined with cascaded lightweight encodings
 - pre-tessellated polygon meshes stored in the tile → skip tessellation on the client
 - up to $3\times$ smaller on uncompressed tiles and $2\text{-}3\times$ faster to decode than MVT

3D TILES

- ▶ OGC standard for streaming and rendering massive 3D geospatial datasets (photogrammetry models , LiDAR, BIM)
- ▶ 3D Tiles uses a concept called Hierarchical Level of Detail to render low-complexity geometry and textures in the distance and high-resolution content nearby
- ▶ Uses glTF as a GPU-friendly payload format, designed for compact representation and minimal runtime processing
- ▶ Powers Google's Photorealistic 3D Tiles, delivering city-scale 3D to third-party apps

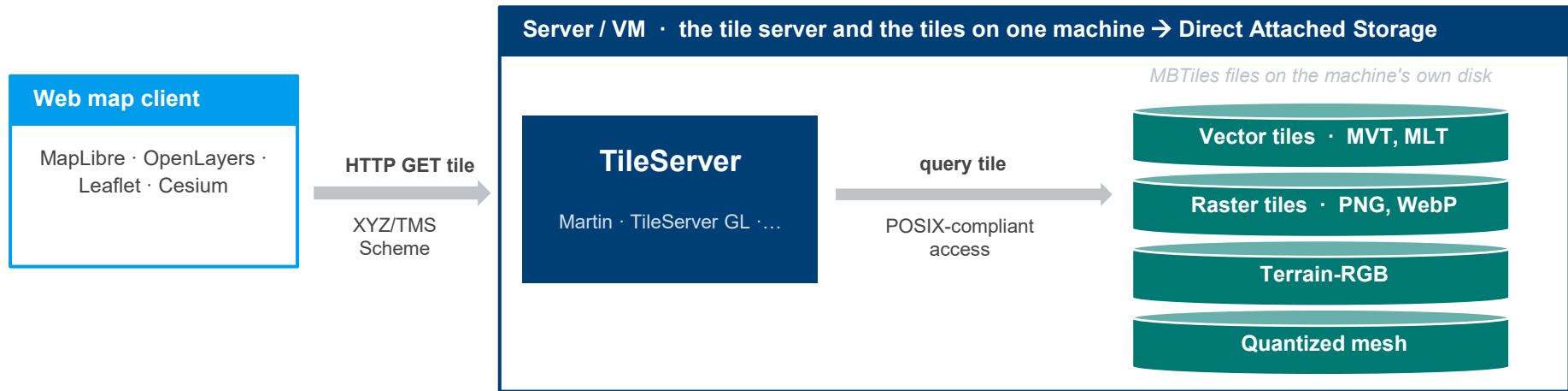


Source: Cesium

Image					
Vertices	~5500	~2880	~1580	~670	140
Notes	Maximum detail, for closeups.				Minimum detail, very far objects.

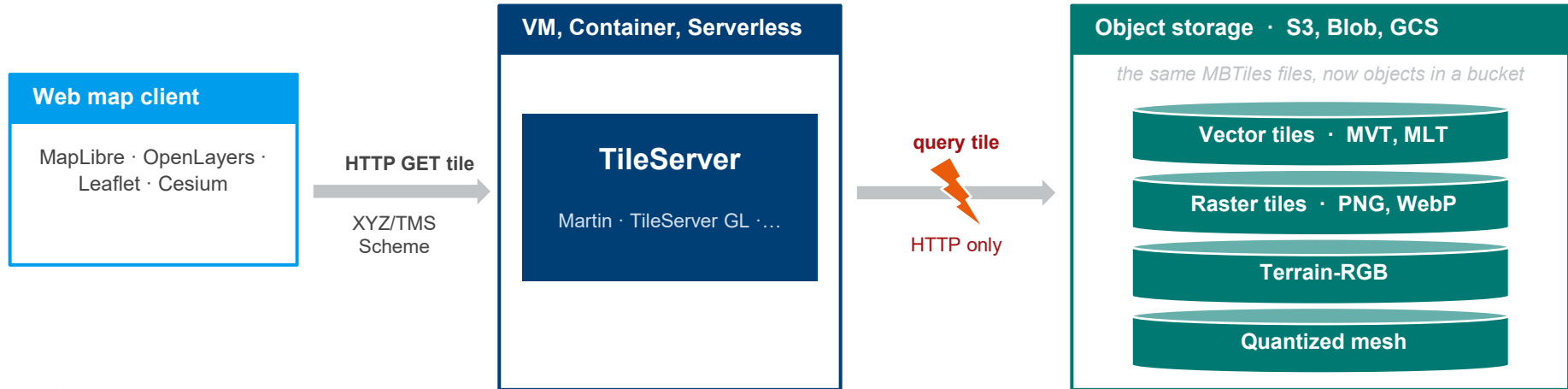
TRADITIONAL TILE ARCHIVES

- ▶ A planet-scale basemap up to zoom 15 is the sum of 4^z tiles $\approx$ 1.4 billion objects
- ▶ Classical tile archives, such as MBTiles and OGC GeoPackage, are based on a SQLite database and enable efficient storage and SQL-based querying of tiles
- ▶ In client-server architectures, an additional tile server is typically used to translate HTTP requests from map clients into SQL queries for accessing tiles stored in the SQLite database



TRADITIONAL TILE ARCHIVES

- ▶ For large geospatial datasets, cloud object storage is preferred due to its cost-efficiency, reliability, and scalability
- ▶ In cloud architectures, however, a basic principle is the separation of storage and compute for scalability reasons
- ▶ Traditional formats rely on assumptions that no longer fit modern data-intensive workflows: co-located storage and compute, and local processing of complete datasets



CLOUD OPTIMIZED FORMATS

- ▶ Cloud-optimized formats support efficient access to selected portions of a file stored in cloud object storage, such as AWS S3, via HTTP Range requests
- ▶ PMTiles and COMTiles package large tile collections for cloud-native distribution without requiring a dedicated tile backend, enabling direct browser access to tiles stored in cloud object storage

Category	Formats
Raster	Cloud Optimized GeoTIFF (COG)
Multi-dimensional raster	Zarr · TileDB
Point cloud	Cloud Optimized Point Cloud (COPC)
Vector	GeoParquet · FlatGeobuf
Tile	PMTiles · COMTiles

Five data types, one design: put an index in the file, and let the client fetch the bytes behind it.

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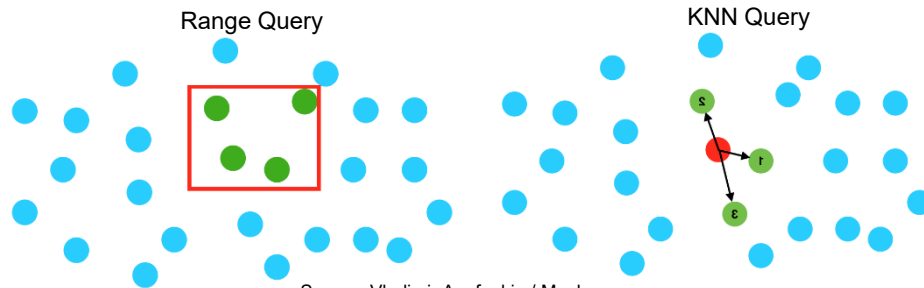
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ANALYTICAL FORMATS

- ▶ Analytics formats are optimized for geospatial analytical workloads, including spatial range queries, k-nearest-neighbor searches, and spatial joins
- ▶ These formats can be further distinguished by whether they are primarily designed for Online Transaction Processing (OLTP) or Online Analytical Processing (OLAP)
- ▶ **GeoPackage**
 - Built on SQLite, providing transactional guarantees and low-latency local access
 - Designed for direct-attached/local storage, not partial or parallel network reads, which limits scalability
 - Best suited for small to medium datasets and common in desktop GIS or map-server workflows such as QGIS and GeoServer



Source: Vladimir Agafonkin / Mapbox

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- ▶ **GeoParquet**: Column-oriented layout on disk
 - Optimized for large-scale OLAP scans
 - Enables Predicate & Projection Pushdown: only read the columns and chunks you need
 - Zone maps (min-max indices) for rapid spatial filtering
 - Lightweight encodings (rle, dictionary, ...) before general compression (Zstd, Snappy)
 - Integrates with modern data analytics ecosystems, including BigQuery, Apache Sedona, DuckDB, and GeoPandas
- ▶ **GeoArrow**: The in-memory counterpart
 - Optimized for CPU throughput and zero-copy access
 - Directly powers modern data science (GeoPandas, DuckDB)

WHAT'S NEXT?

Visualization

- ▶ **Cartographic Immersive 3D map views**: symbolized city views instead of photorealism → MLT 3D
- ▶ **Gaussian splatting** is arriving in 3D Tiles pipelines as a payload alongside glTF meshes
- ▶ **Learned geometry compression** based on Machine learning approaches

Analytics

- ▶ Parquet's limitations are being addressed by newer formats such as **FastLanes, Nimble, Vortex, Lance and BtrBlocks**
- ▶ Specialized **cloud-native index structures**
- ▶ File formats for **spatial embeddings**
- ▶ **Routing support** insided vector tiles MLT

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Cartography-Oriented 3D Visualization (Symbolised 3D Models)
Examples: Google Maps Immersive Navigation, Apple Maps Detailed City Experience



Source: Apple Maps screenshot © Apple Inc.

Photorealistic Rendering
(not a primary focus, as this is already well supported by 3D Tiles)
Examples: Google Maps Photorealistic 3D Tiles

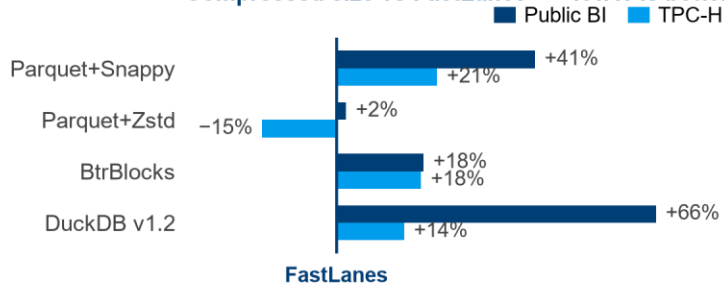


Source: Apple Maps screenshot © Apple Inc.

Screenshots: Apple Maps © Apple Inc.

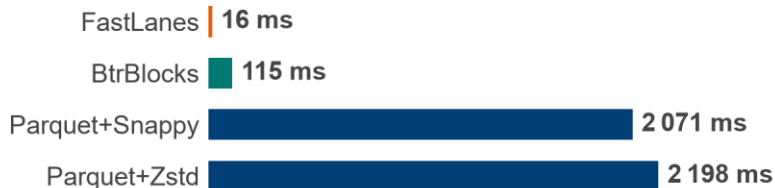
WHAT'S NEXT?

Compressed size vs FastLanes · lower is better



Only Parquet+Zstd matches FastLanes on size — and it is the slowest to decode.

Total decode time · lower is better



Public BI on Intel Ice Lake, AVX-512. That build gains FastLanes 42%, Parquet+Snappy 2%.

Source: FastLanes Paper

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THANK YOU — QUESTIONS?

Markus Tremmel

Rohde & Schwarz

Happy to argue about any box in the taxonomy — especially the ones you think are in the wrong column.